ENGINEERING DIAGRAMS · MICROSOFT 365 EDITION · OCTOBER 2026

System design diagrams

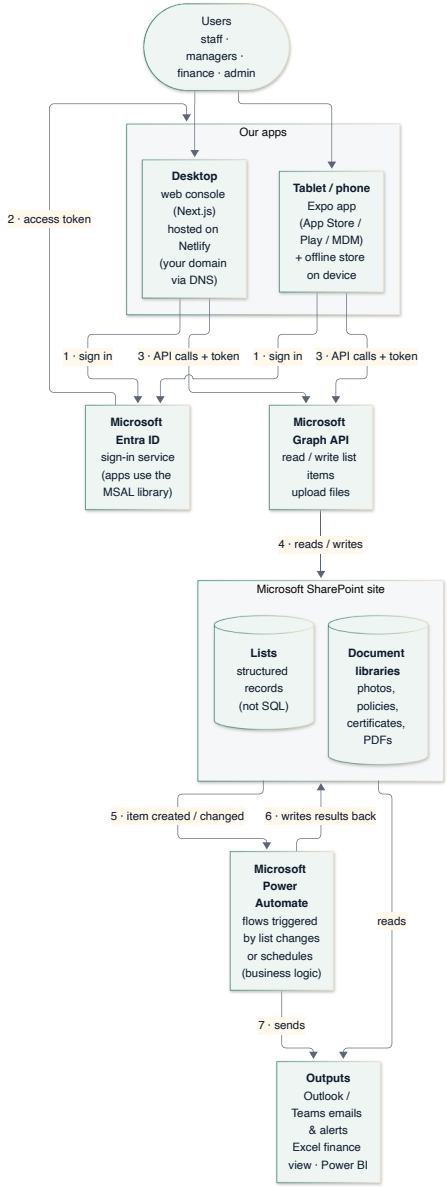
Architecture, security, flows, lifecycles, sequences, SharePoint data model, scalability and roadmap for the restaurant operations apps built on Maginhawa's Microsoft 365 (SharePoint + Power Automate + Entra ID). Companion to the Tech stack and architecture plan.

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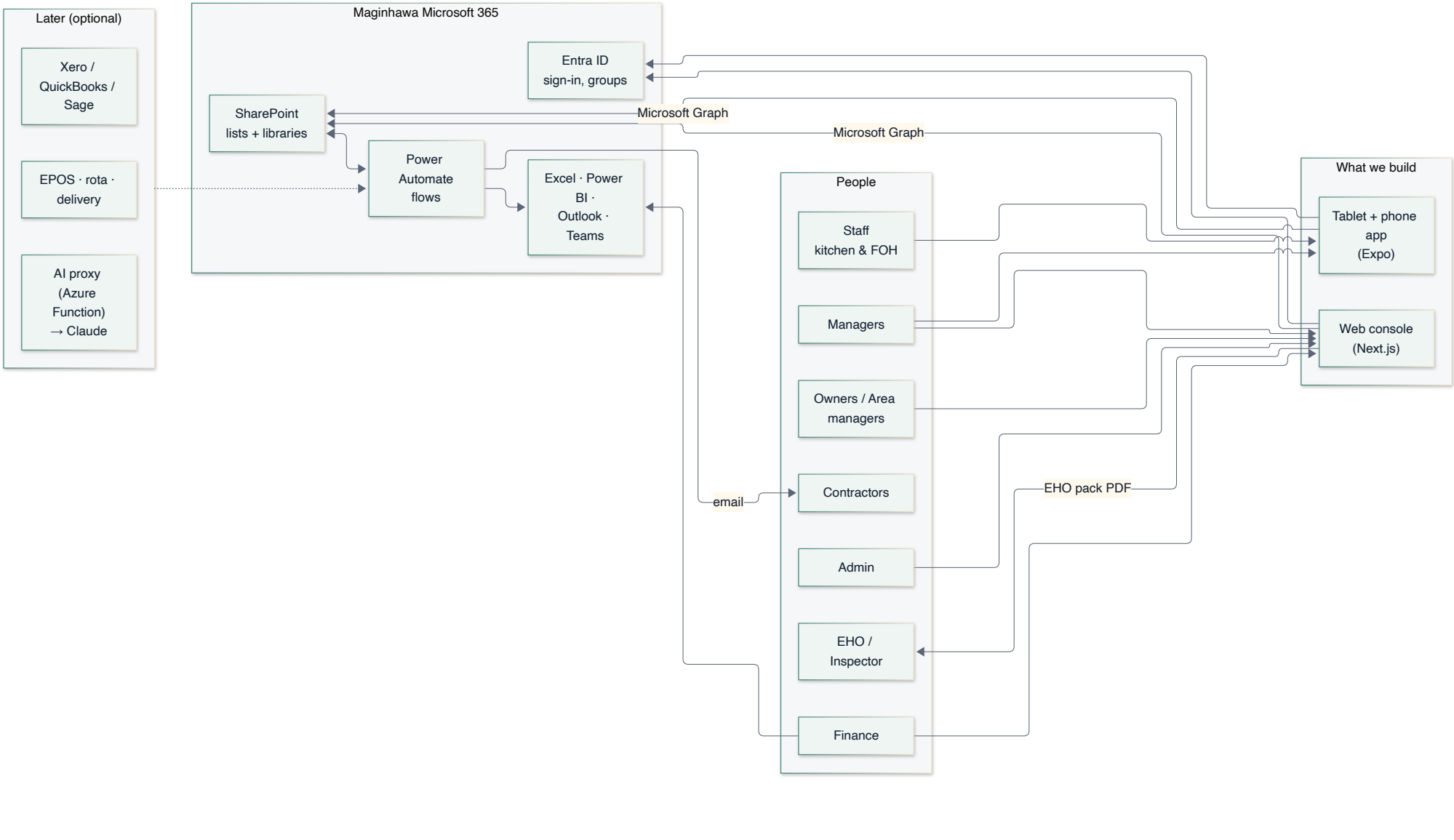
The simple picture

Our apps sign in through Entra ID (MSAL), call the Microsoft Graph API with the token, and Graph reads and writes SharePoint. Power Automate sits beside SharePoint: flows start when items change or on a schedule, write results back, and send emails. Excel and Power BI read SharePoint.



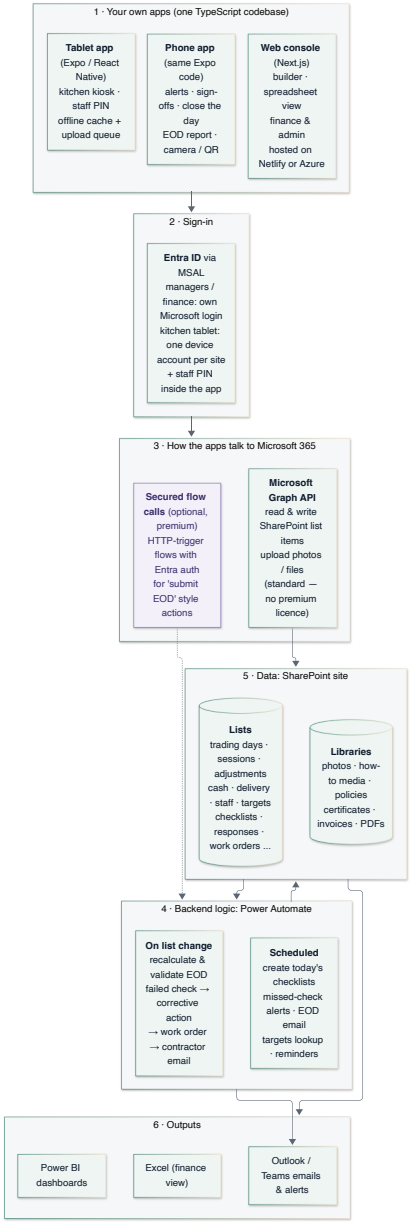
System context

Who uses the system, what we build (apps and console), what Maginhawa's Microsoft 365 provides, and the optional later integrations.



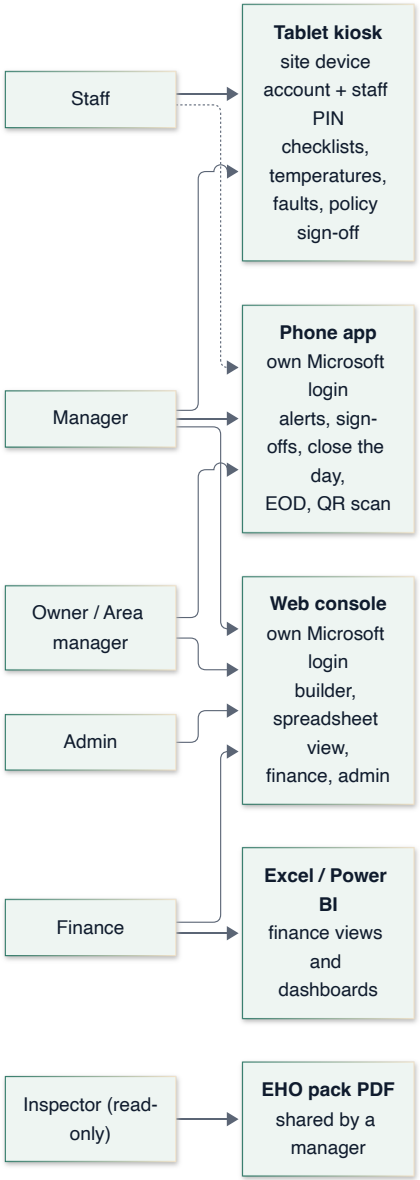
Architecture in layers

Our apps → Entra sign-in → Graph (standard) or optional secured flow calls (premium) → SharePoint lists and libraries ↔ Power Automate → outputs.



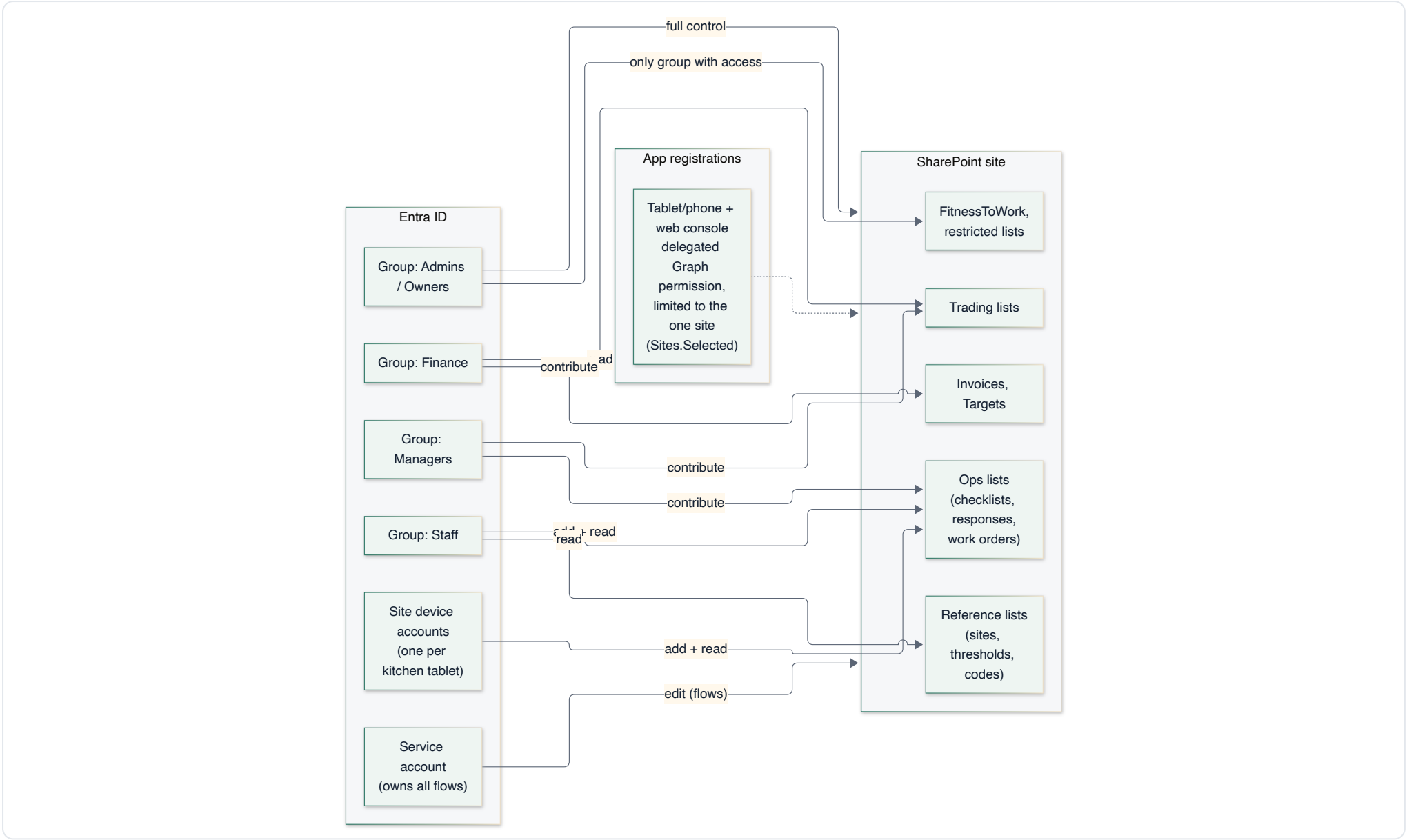
Who uses which device

Kitchen tablet on a site device account with staff PINs; phone and web console with personal Microsoft logins; finance in Excel / Power BI; inspectors receive the EHO pack PDF.



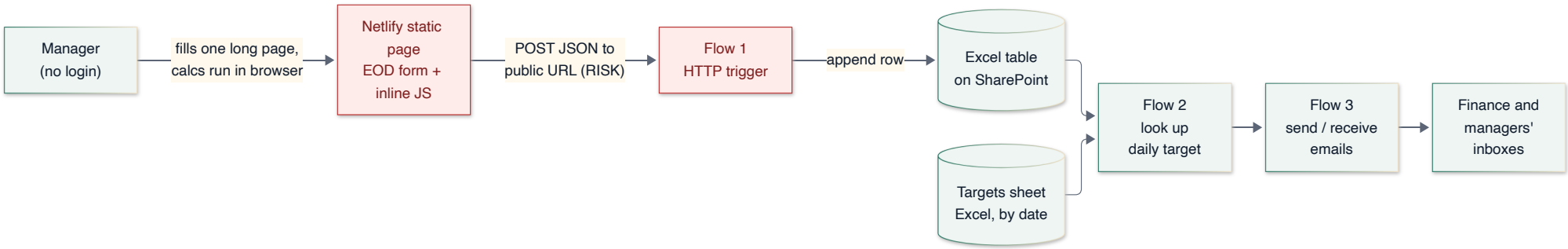
Security and access

Entra groups map to SharePoint permissions. The device accounts and staff can add and read operational lists; finance has its own lists; restricted lists are for Owners/Admins only; the service account owns every flow. The app registration is limited to the one site.



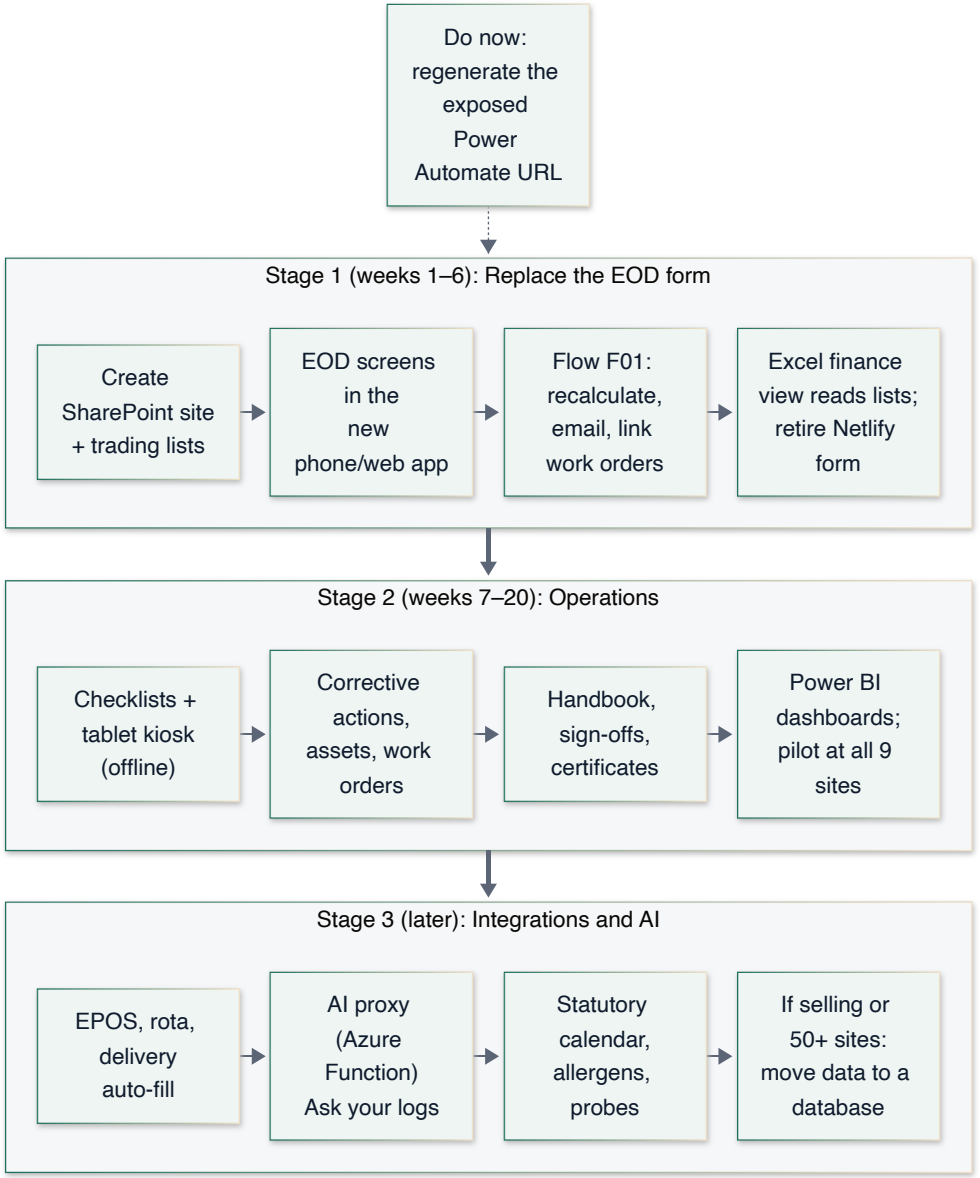
Current EOD system (as-is)

Today's Netlify form: browser-only calculations, a public Power Automate URL (red = risk), a row appended to Excel, then the targets lookup and emails.



Migration plan

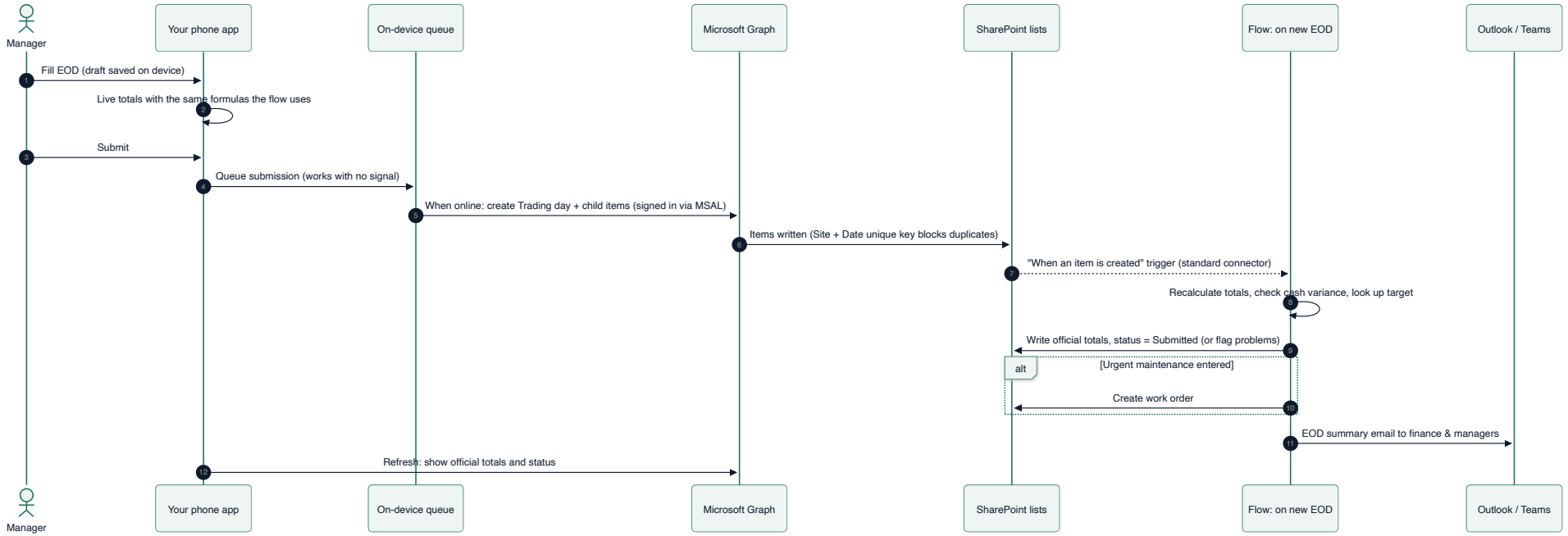
Stage 1 replaces the EOD form within weeks (SharePoint lists, flow F01, Excel view, Netlify retired). Stage 2 adds operations. Stage 3 adds integrations and AI, and moves to a database only if selling or 50+ sites.



FINANCE EOD · 8/21

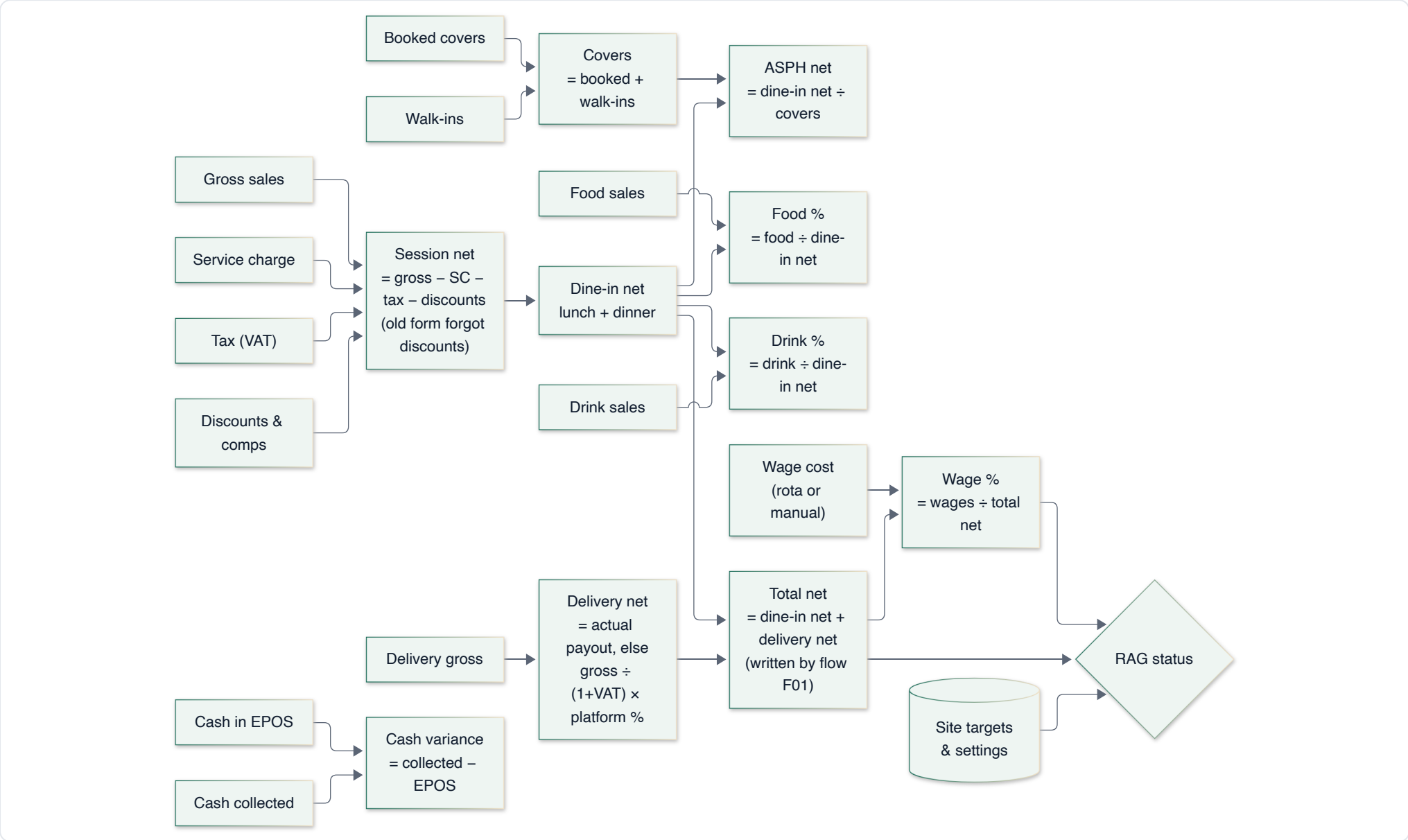
EOD submission, step by step

The app saves a draft and a preview on the device, queues and uploads through Graph, and the unique Site + Date key blocks duplicates. Flow F01 recalculates, checks, writes official totals, creates a work order for urgent maintenance and emails finance.



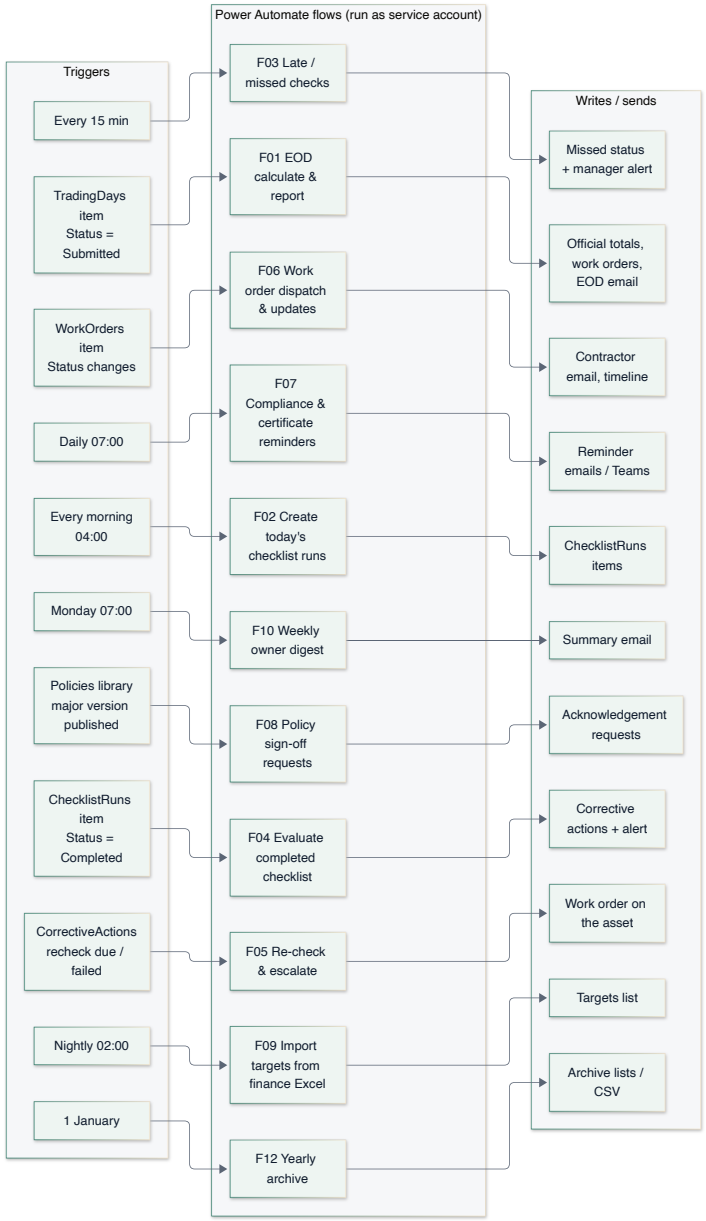
Trading calculations (corrected)

How each EOD figure is derived, with the old form's bugs fixed: discounts are subtracted from net, delivery and wage settings come from lists, and the official totals are written by flow F01.



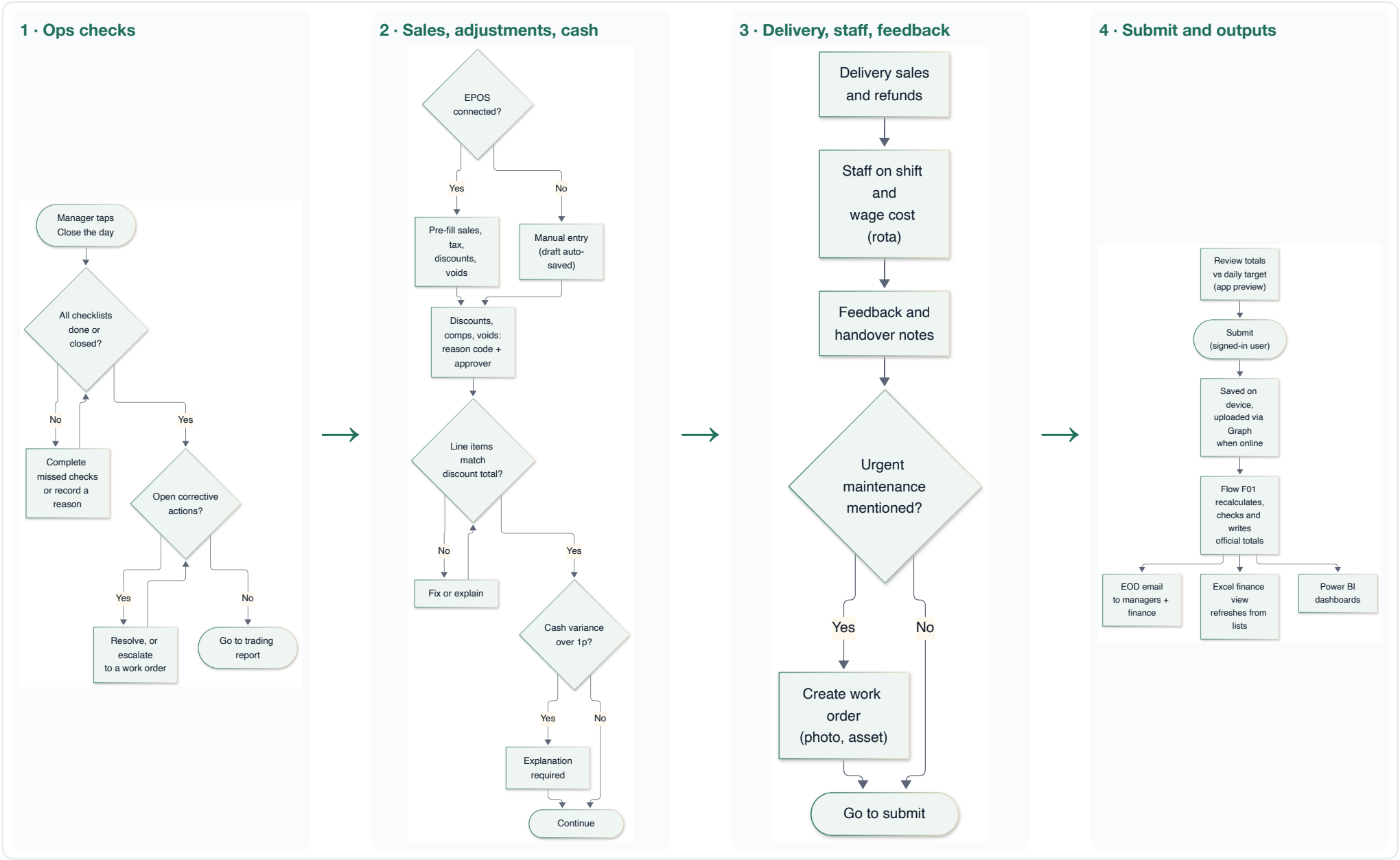
Power Automate flow catalogue

Eleven flows, F01 to F12, with their triggers and what they write or send. All are owned by the service account and use standard connectors.



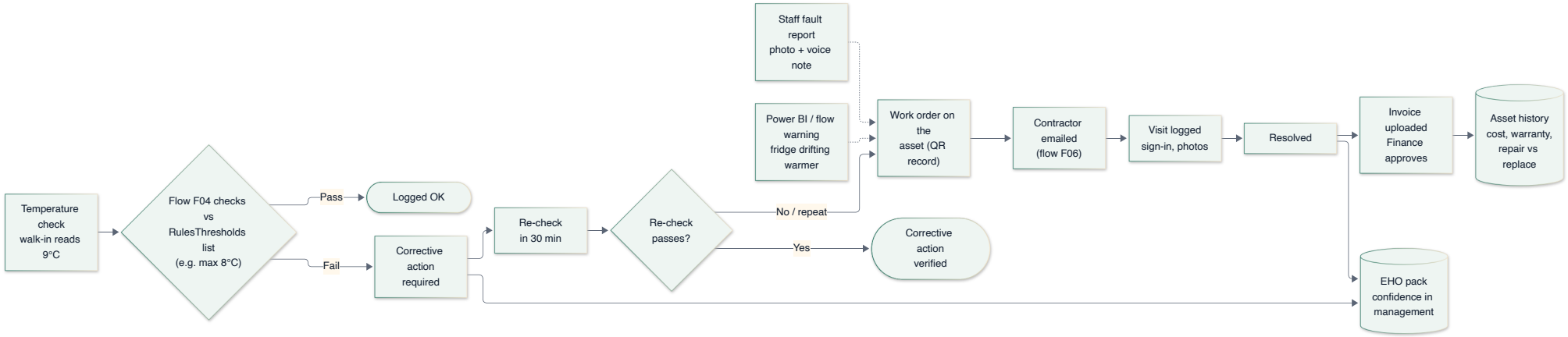
Close-the-day flow (manager)

Ops checks, then sales, adjustments and cash, then delivery, staff and feedback, then submit and outputs.



The closed loop: failed check to repair cost

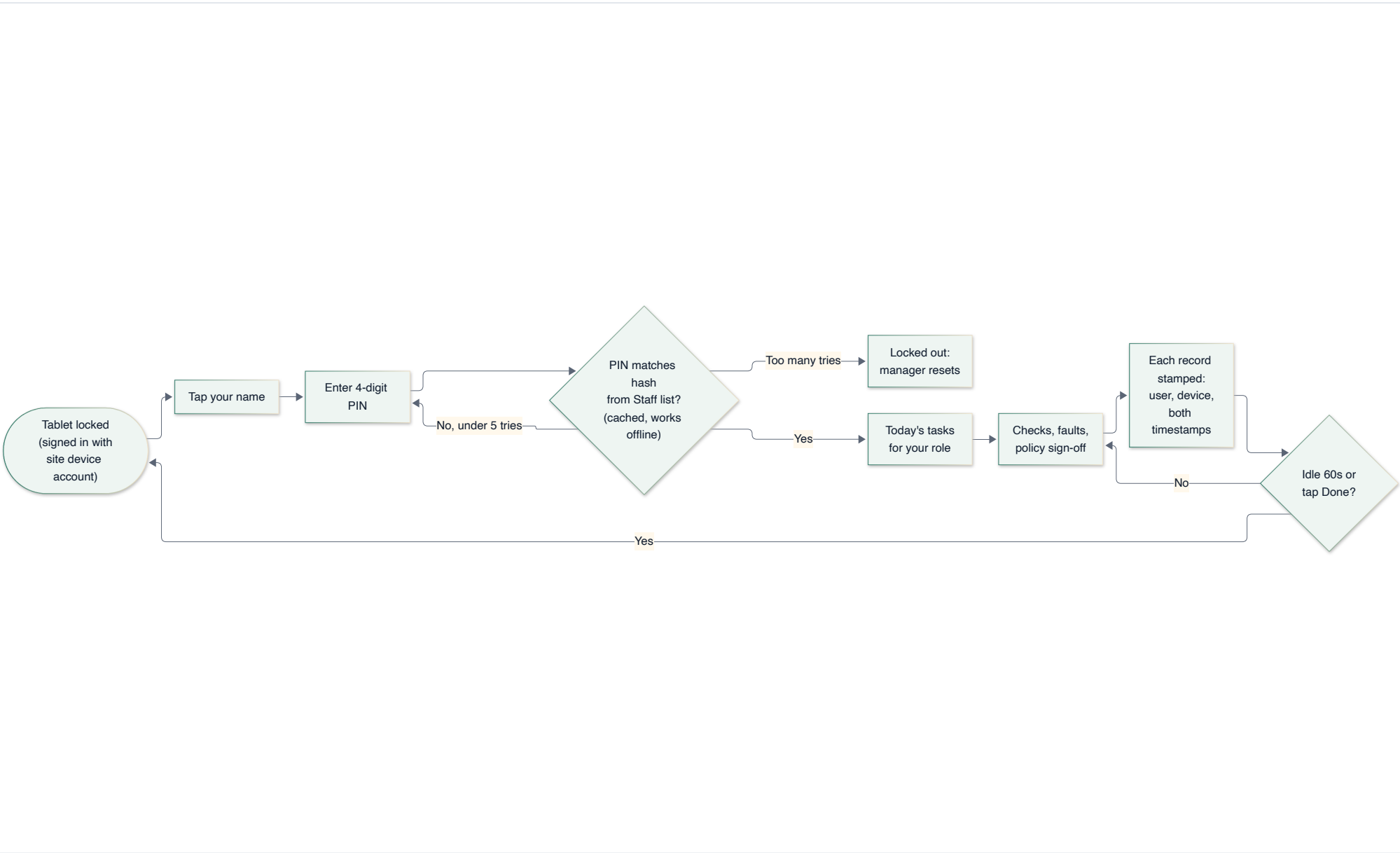
Flow F04 checks the completed checklist against RulesThresholds. A failure forces a corrective action and a re-check (F05); repeats escalate to a work order on the asset, then the contractor email (F06), the resolution, the invoice and the asset's cost history, all feeding the EHO pack.



FLOWS · 13/21

Shared tablet: PIN login

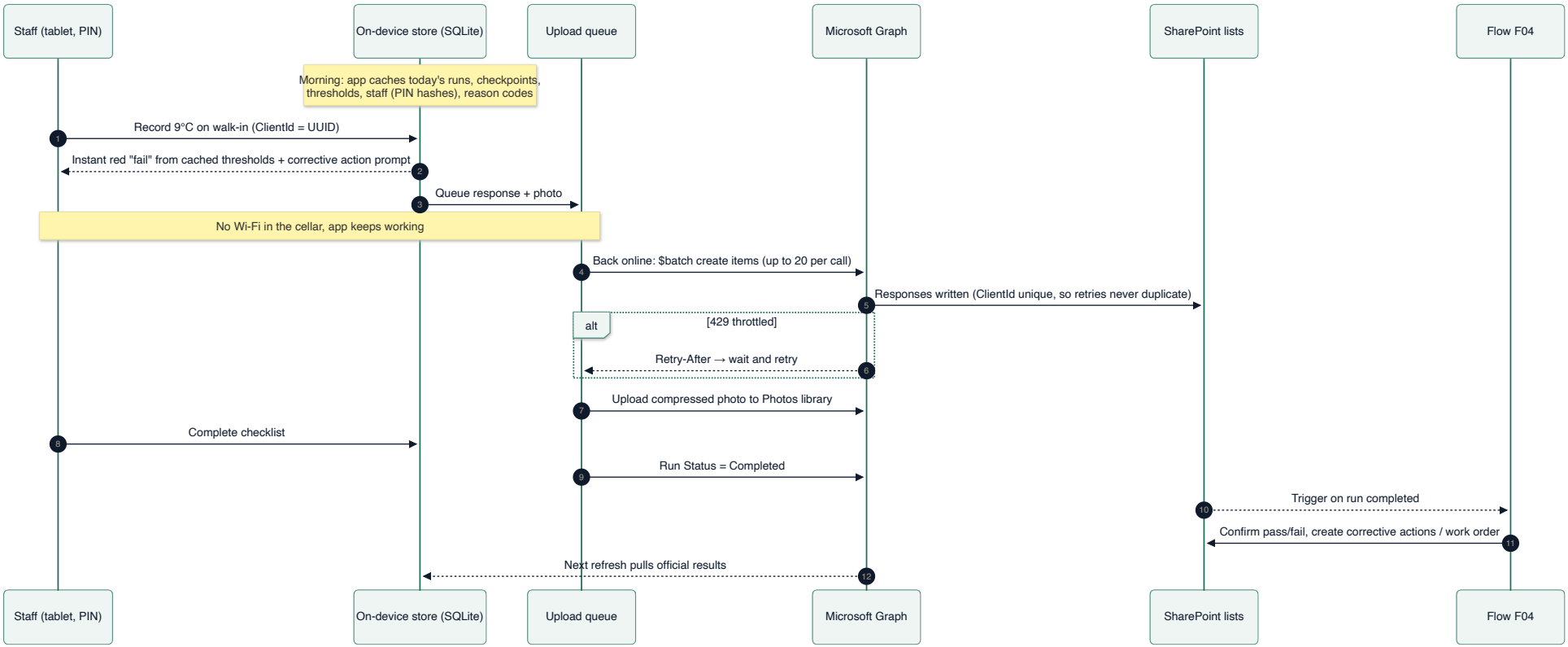
The tablet stays signed in with the site device account. Staff pick their name and enter a PIN checked against the cached Staff list (works offline). Every record is stamped with the person, the device and both timestamps.



FLOWS · 14/21

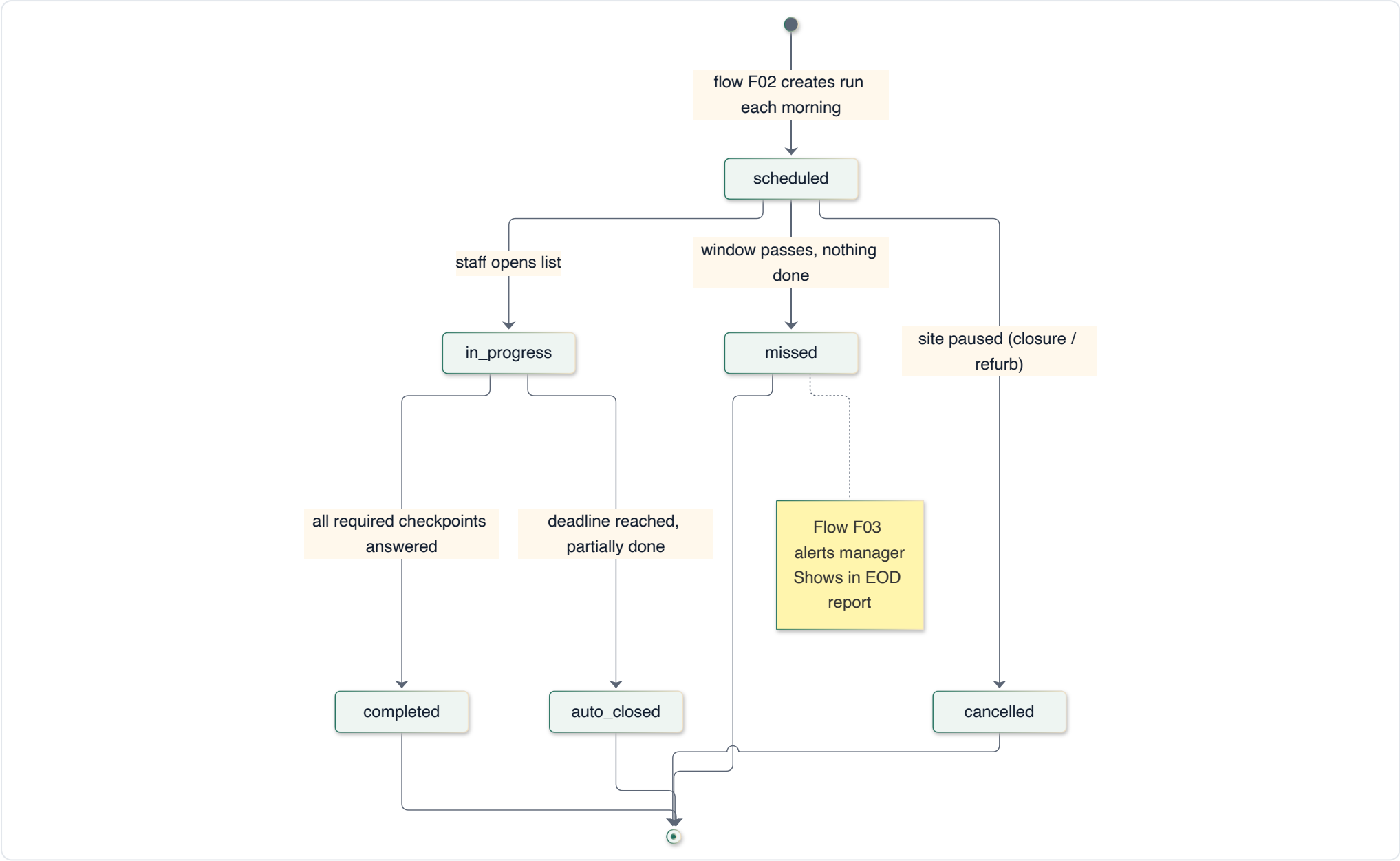
Offline capture and sync through Graph

Morning cache, instant local pass/fail, queued $batch uploads with ClientId so retries never duplicate, Retry-After on throttling, background photo upload, then flow F04 confirms the results.



Checklist run lifecycle

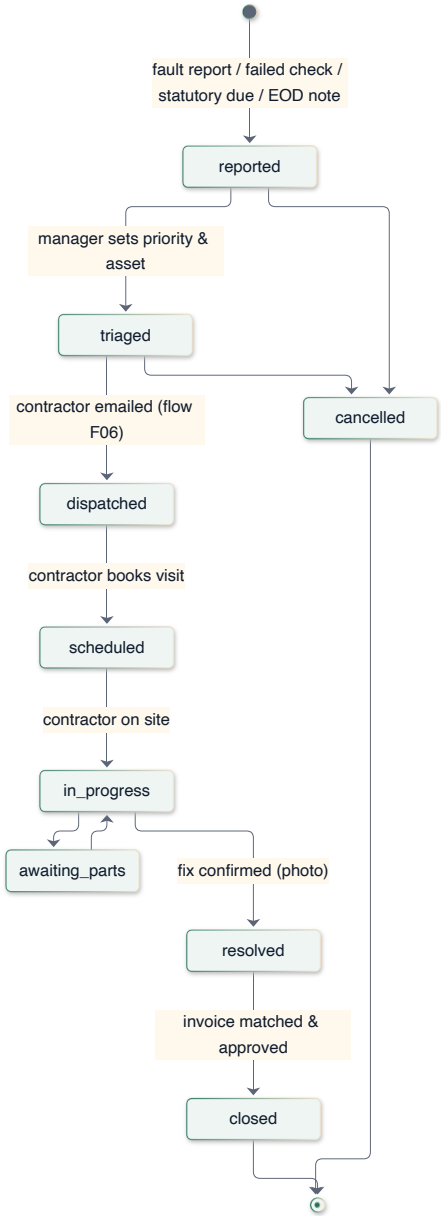
Created each morning by flow F02, then completed, missed (alert from F03), auto-closed or cancelled during closures.



LIFECYCLES · 16/21

Work order lifecycle

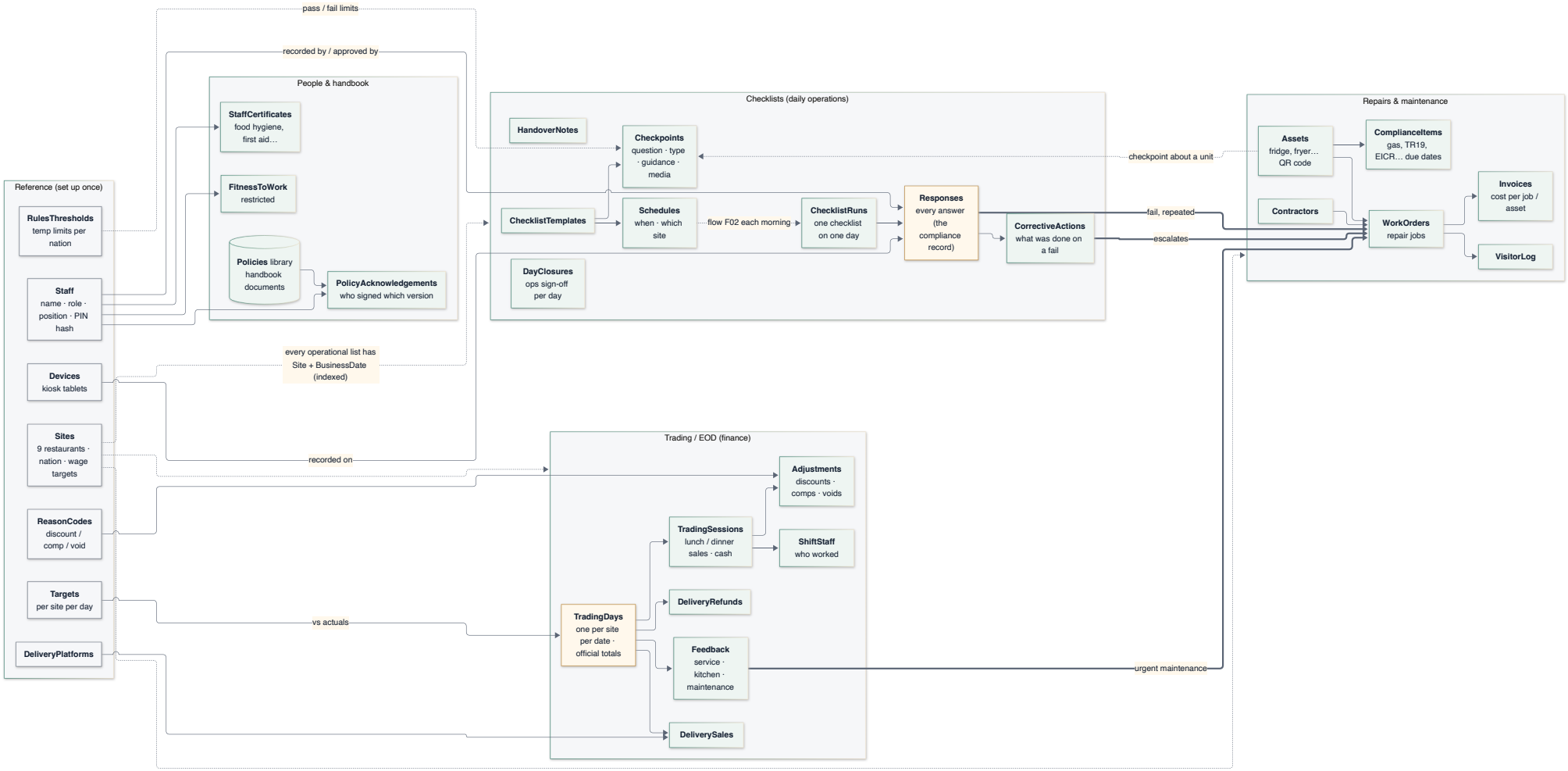
From reported to closed. 'Closed' requires the invoice to be matched and approved.



DATA MODEL

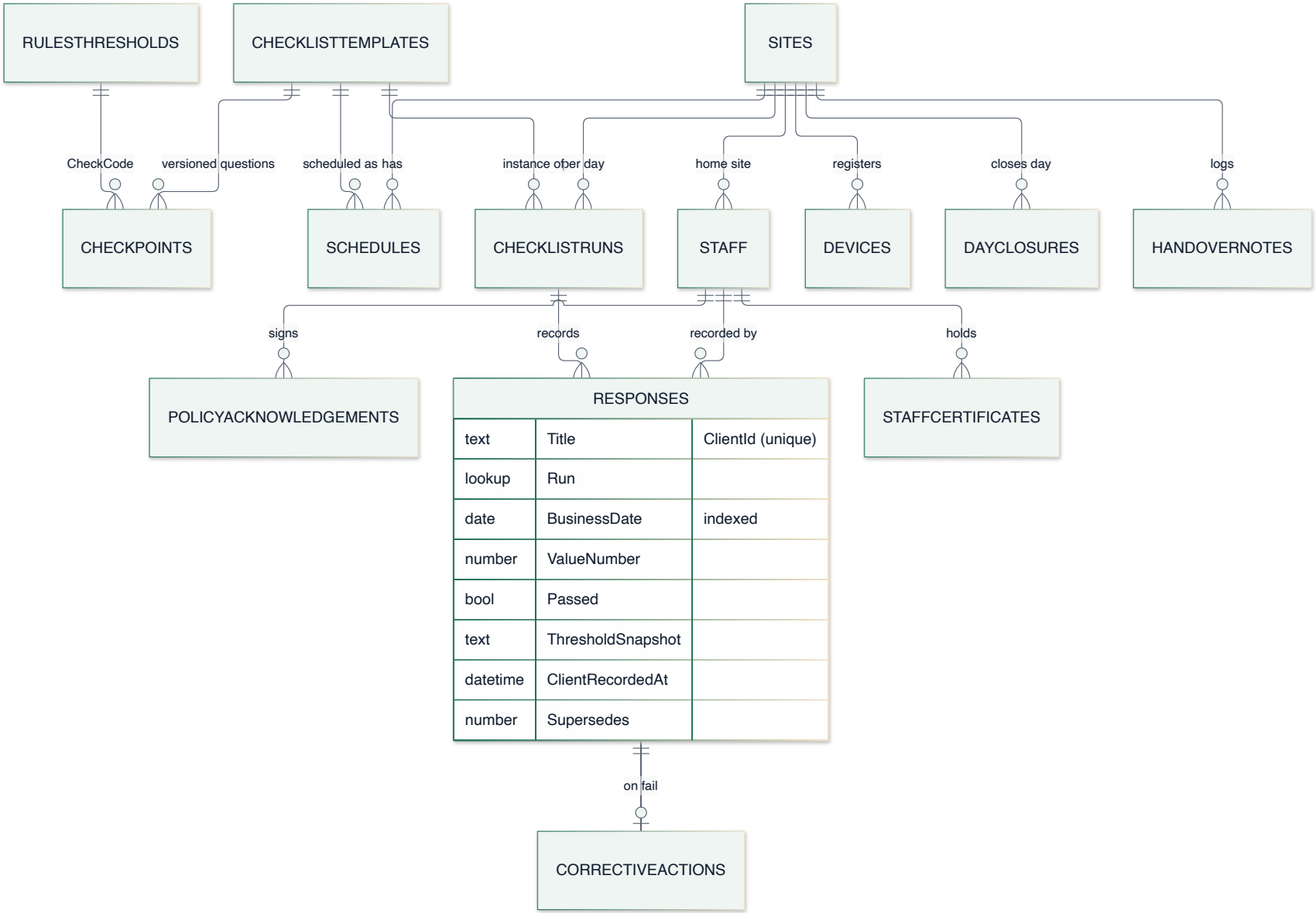
Data model overview: all 31 lists

Five groups. Reference data is set up once. Checklists create one Run per checklist per day, and every answer is a Response. Failed responses, corrective actions and EOD maintenance notes all lead to WorkOrders on an Asset. Each TradingDay holds lunch and dinner sessions with their line items. Every operational list carries Site + BusinessDate (indexed). Orange = highest-volume lists.



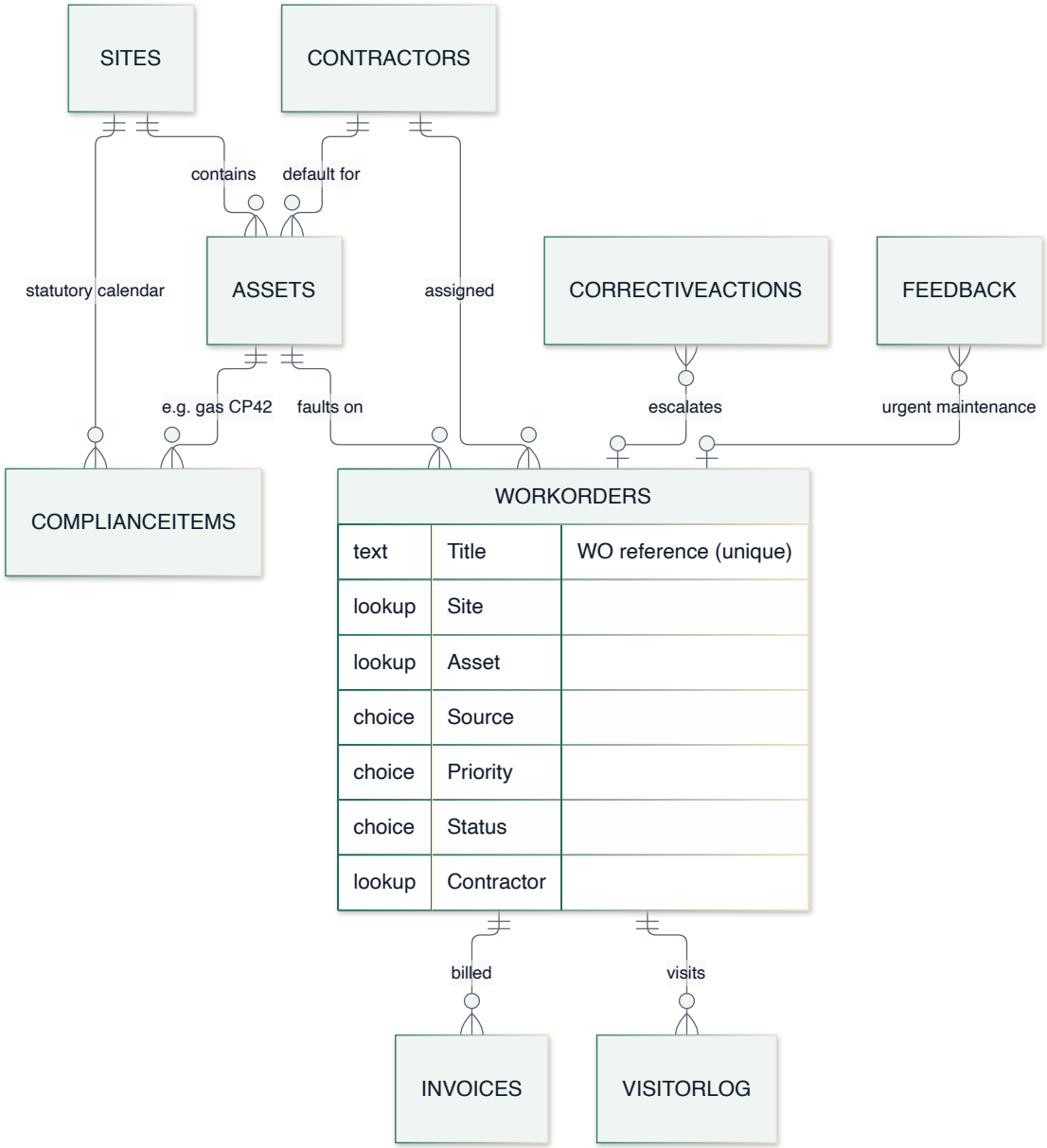
SharePoint lists: operations

Sites, staff, devices, versioned checklist templates and checkpoints, schedules, runs, responses (the compliance record), corrective actions, day closures, sign-offs and certificates.



SharePoint lists: maintenance

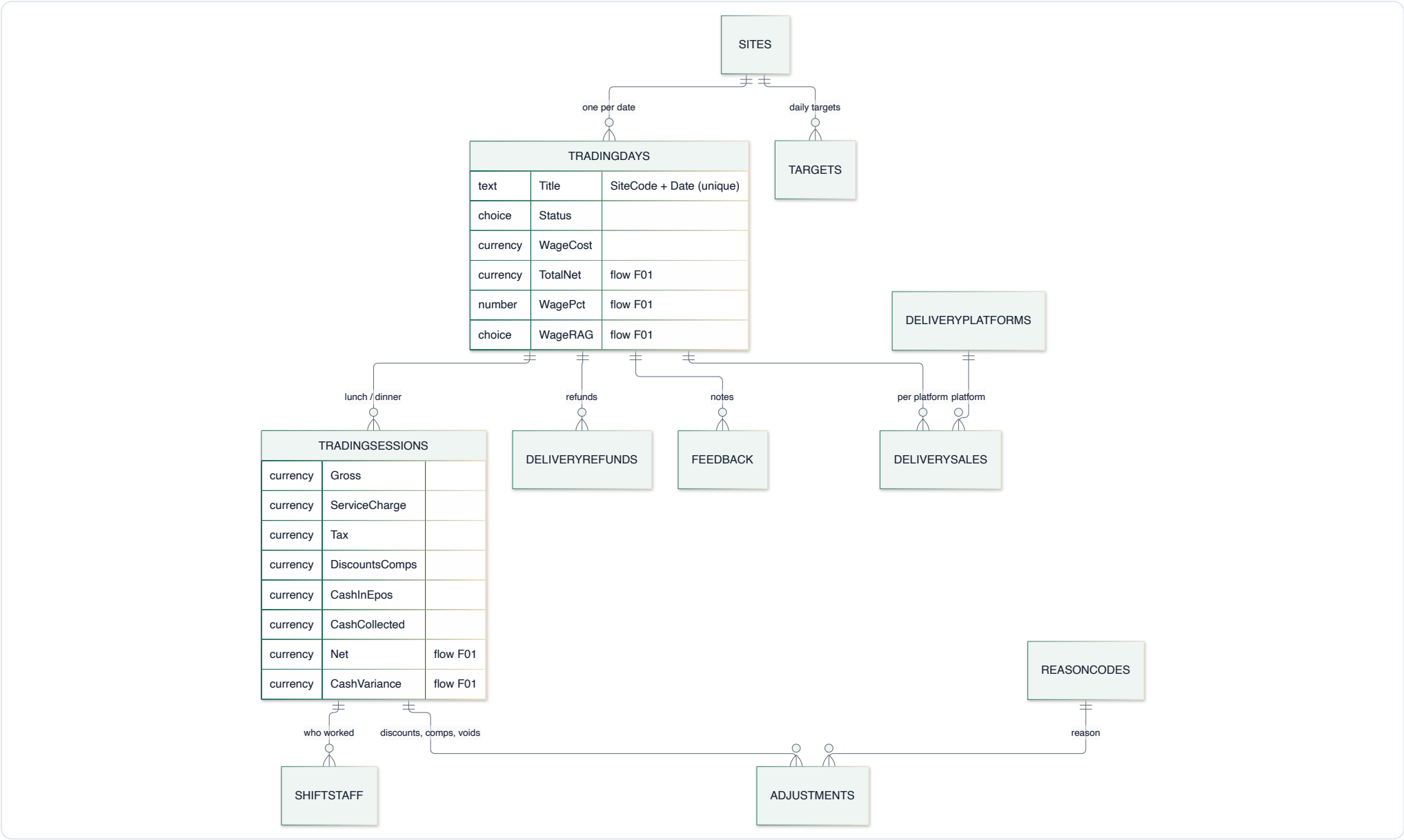
Assets, contractors, work orders (from failed checks, fault reports, EOD notes), the statutory compliance calendar, invoices and the visitor log.



DATA MODEL · 19/21

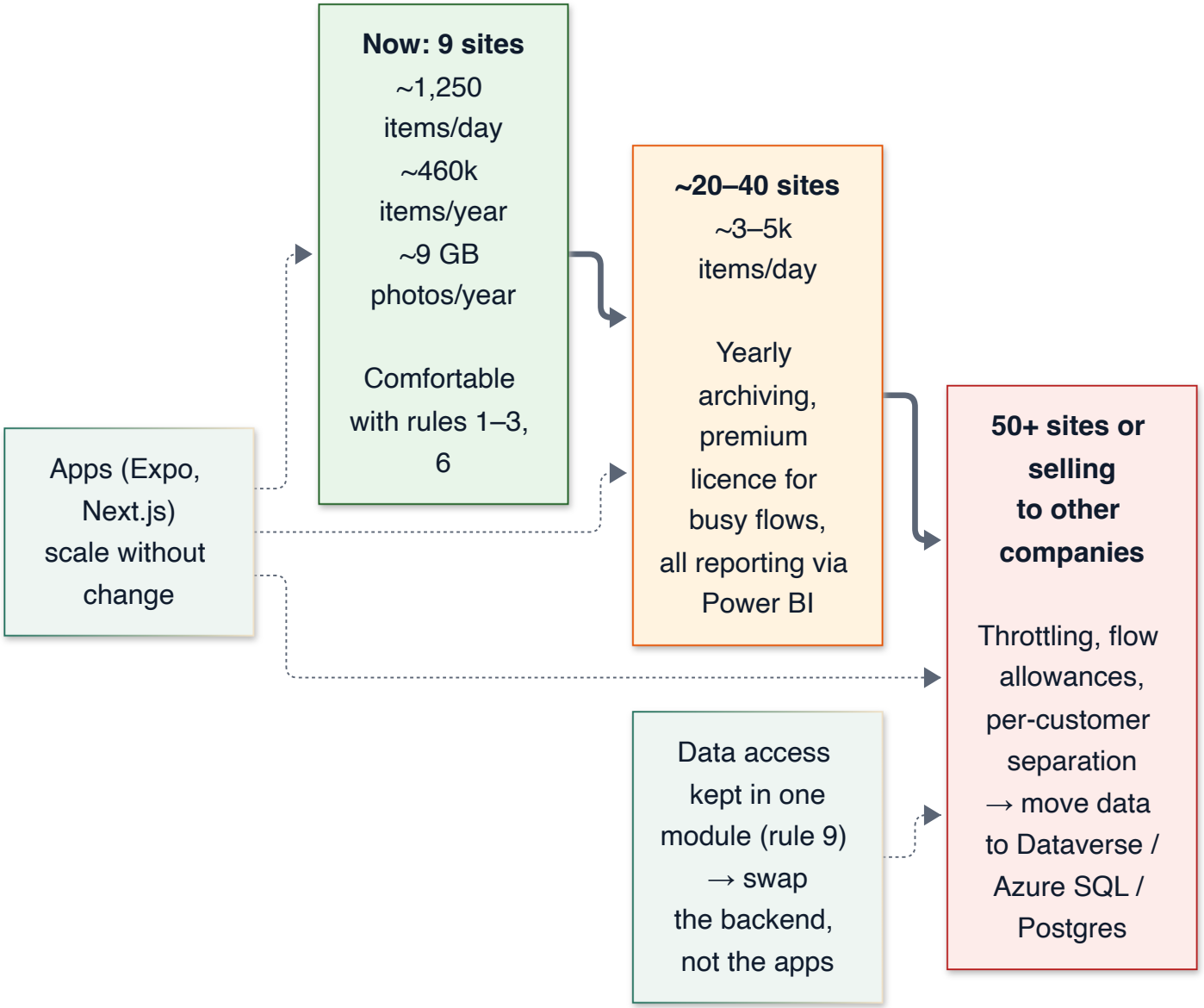
SharePoint lists: trading / EOD

Trading days and sessions, adjustments with reason codes, delivery sales and refunds, shift staff, feedback and targets. Columns marked 'flow F01' are written only by the flow.



Scalability tiers

Comfortable at 9 sites; workable at about 20–40 sites with archiving, licences and Power BI; at 50+ sites or when selling, move the data to a database. The apps don't change.



Build roadmap

Setup, Stage 1 EOD (retire Netlify), Stage 2 operations, hardening and a pilot at all 9 sites, then Stage 3 integrations and AI.

